

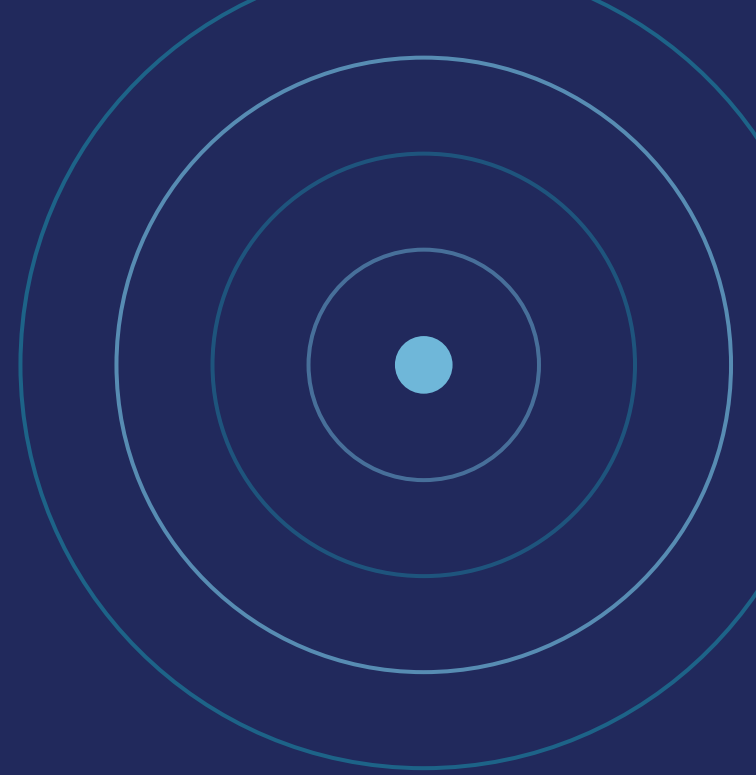
CA4102 · MCA 1ST YEAR · GEETA UNIVERSITY

Introduction to TCP/IP

Lecture 7 & 8: Introduction and History of TCP/IP; Layers of TCP/IP and Protocols

Instructor: Dr. Shamneesh Sharma

Lecture Dates: 20 Aug 2026 (Thursday)



Today's Roadmap

7

Lecture 7 — History of TCP/IP

- What Is TCP/IP?
- Why TCP/IP Was Needed: Design Goals
- TCP/IP Timeline: From ARPANET to the Internet
- Cerf & Kahn and the Original 1974 Paper
- The 1983 "Flag Day"
- Why TCP/IP Became the Global Standard

8

Lecture 8 — Layers & Protocols

- The TCP/IP Protocol Suite: Layer Overview
- TCP/IP vs OSI: Layer Mapping
- Physical & Data Link Layers
- Network (Internet) Layer: IP, ARP, ICMP
- Transport Layer: TCP, UDP, SCTP
- Application Layer & Addressing

Learning Outcomes

Course Outcome CO2 (Bloom's Level: Create) — Design appropriate protocol-based architectures for a given networking scenario, using the TCP/IP suite as the underlying framework.

By the end of these two lectures, students will be able to:

- 1 Explain what the TCP/IP protocol suite is and why it was developed
- 2 Trace the historical evolution of TCP/IP from ARPANET/NCP to the modern Internet standard
- 3 Justify why TCP/IP became the dominant protocol suite over the competing OSI stack
- 4 Describe the layers of the TCP/IP protocol suite and map them to the OSI model
- 5 Identify the core protocols operating at each TCP/IP layer and design solutions using them

What Is TCP/IP?

TCP/IP is not a single protocol but a **protocol suite** — a family of communication protocols that together define how data is addressed, transmitted, routed, and received across interconnected networks.

- Named after its two foundational protocols: Transmission Control Protocol (TCP) and Internet Protocol (IP)
- Defines a common language so dissimilar networks and devices can interoperate
- Forms the technical foundation of the modern Internet
- Organized as a set of layers, each contributing a specific piece of the communication task

Why It Matters

Every device that connects to the Internet — a laptop, a phone, a server, a router — communicates using TCP/IP.

It is the reason a smartphone in India can exchange data with a server in the USA, despite running on completely different underlying hardware.

Why TCP/IP Was Needed: Design Goals

1

Robustness

The network must continue operating even if individual links or nodes fail — no single point of failure

2

Heterogeneity

Must support many different underlying network technologies — radio, satellite, cable — communicating seamlessly

3

Distributed Control

No central point manages the whole network; control and routing decisions are distributed

4

Resource Efficiency

Use packet switching to share network capacity efficiently among many simultaneous conversations

5

Modularity

A layered, modular design so pieces can evolve independently over time

6

Accountability

Provide a way to attribute and manage usage of network resources across multiple organizations

TCP/IP — Key Milestones

1969

ARPANET launched using NCP
(Network Control Protocol)

1978

TCP is split into two protocols:
TCP and IP

1983

TCP/IP bundled into 4.2BSD
Unix — free & widely available

1974

Vint Cerf & Bob Kahn publish
the TCP design

1983

"Flag Day": ARPANET switches
from NCP to TCP/IP

1990s

NSFNET & commercial ISPs
drive global TCP/IP adoption

Cerf & Kahn and the Original TCP Paper

- In May 1974, Vinton Cerf and Robert Kahn published "A Protocol for Packet Network Intercommunication" in IEEE Transactions on Communications
- The paper proposed a single, unifying protocol to interconnect independent packet-switched networks — the birth of "internetworking"
- Their design introduced gateways (now routers) to relay packets between otherwise incompatible networks
- For this contribution, Cerf and Kahn are widely credited as the "fathers of the Internet"
- Their original protocol was later split (1978) into TCP (reliable delivery) and IP (addressing & routing)

Why Splitting TCP & IP Mattered

IP: handles addressing and best-effort routing of individual packets.

TCP: handles reliability — ordering, retransmission, and flow control — on top of IP.

Separating these concerns let applications choose TCP's reliability, or trade it for speed with a lighter protocol (UDP) when reliability wasn't essential.

The 1983 "Flag Day"

January 1, 1983 — every host on ARPANET was required to switch, on the same day, from the old NCP protocol to TCP/IP. This coordinated, all-at-once cutover is remembered as the "Flag Day," and is often marked as the technical birth of the modern Internet.

A Risky, Coordinated Switch

Hosts that failed to switch in time were simply cut off from the network — there was no gradual transition period.

Why the Switch Was Necessary

NCP could not support internetworking across multiple, independently administered networks the way TCP/IP could.

Lasting Impact

The Flag Day proved TCP/IP could operate at scale, paving the way for its adoption well beyond ARPANET.

Why TCP/IP Became the Global Standard

Both TCP/IP and the OSI protocols were developed around the same era — but only one became the world's networking standard.

	TCP/IP	OSI Protocols
Availability	Free, working implementations shipped with Unix by the early 1980s	Formal standard finalized late; implementations lagged
Philosophy	"Rough consensus and running code" — practical, iterative	Exhaustive committee-driven design before deployment
Simplicity	Fewer layers, simpler to implement and debug	7 layers with some overlapping responsibilities
Momentum	Already running the growing ARPANET/NSFNET/Internet	Lacked an existing large-scale network to build on



QUICK CHECK

Quiz 1 — History of TCP/IP

Q1. Which protocol did ARPANET use before switching to TCP/IP?

A) FTP B) NCP C) HTTP D) SNMP

Answer: B) NCP

Q2. In what year did TCP split into the separate TCP and IP protocols?

A) 1969 B) 1974 C) 1978 D) 1983

Answer: C) 1978

LECTURE 9

Layers of TCP/IP and Protocols

Network Access · Internet · Transport · Application

The TCP/IP Protocol Suite — Layer Overview

5

Application

Interfaces with end-user software & services

4

Transport

End-to-end delivery between processes

3

Network (Internet)

Logical addressing & routing across networks

2

Data Link

Framing & physical (MAC) addressing on one link

1

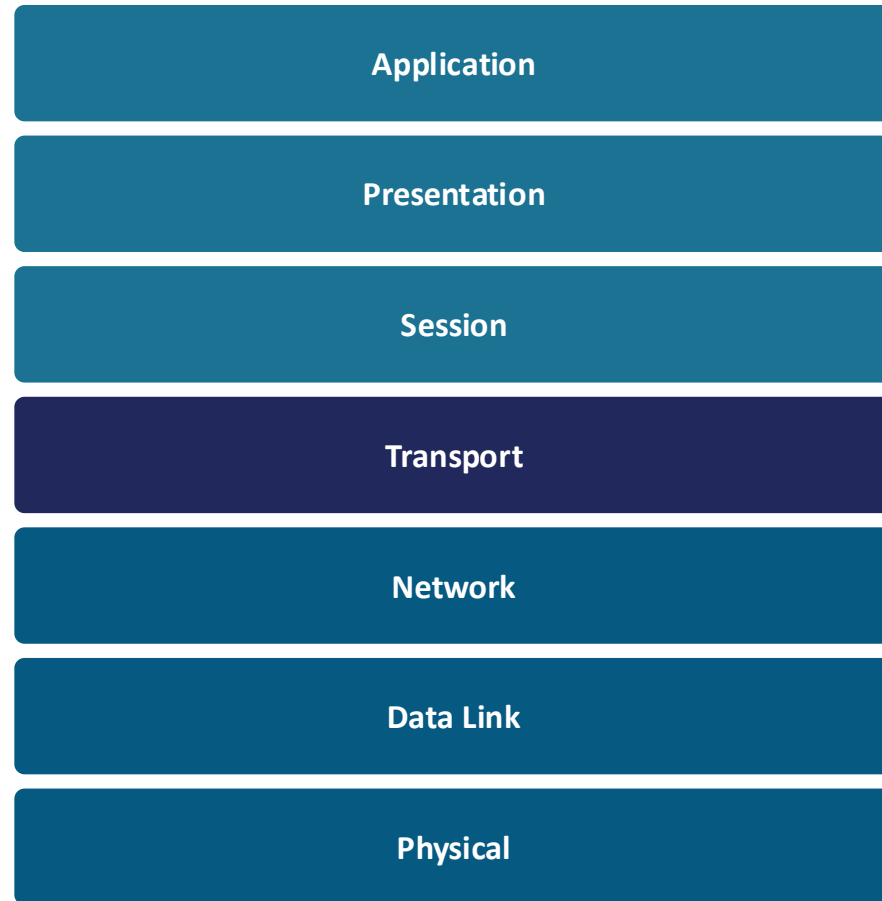
Physical

Raw bit transmission over the medium

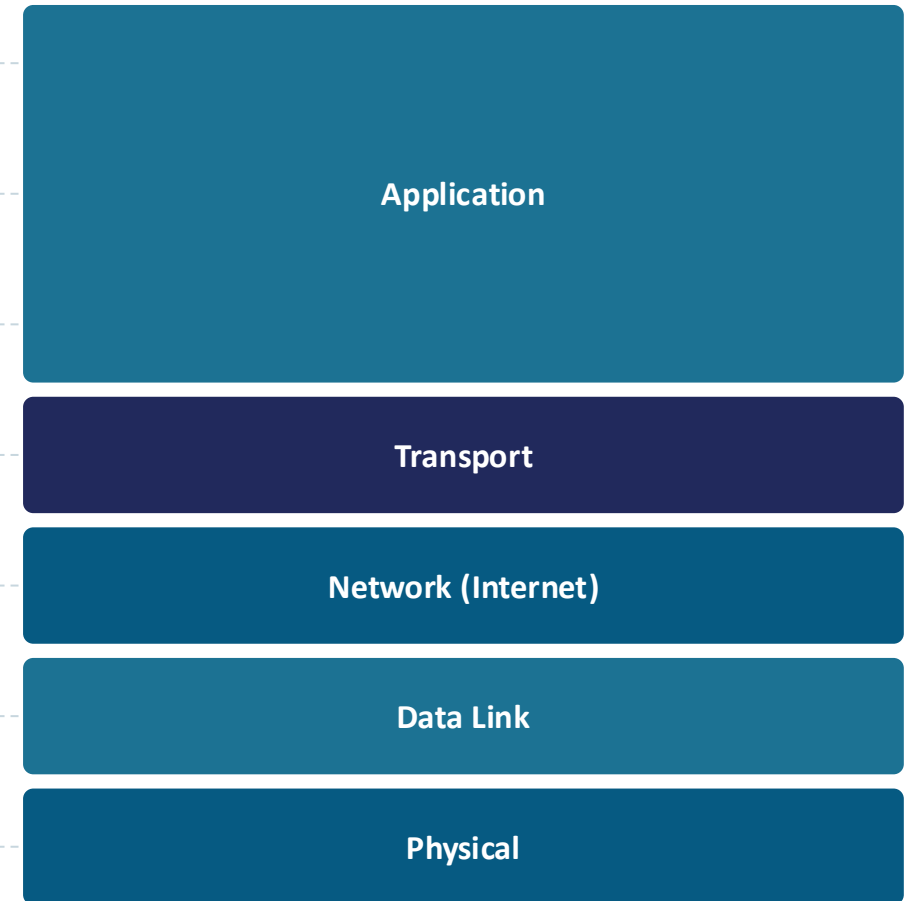
Note: some textbooks combine Physical + Data Link into a single "Network Access / Link" layer, giving a 4-layer model.

TCP/IP vs OSI — Layer Mapping

OSI Model (7)



TCP/IP Suite (5)



Physical & Data Link Layers

Functions & Services

- Physical Layer: transmits raw bits as electrical, optical, or radio signals over the transmission medium
- Data Link Layer: organizes bits into frames and adds physical (MAC) addressing
- Together, these layers are sometimes grouped as the "Network Access" or "Link" layer in the 4-layer TCP/IP model
- Responsible for delivering data correctly across a single physical link — not across the whole internetwork
- The specific technology used here depends entirely on the underlying network (wired, Wi-Fi, cellular, etc.)

Key Protocols

- Ethernet / IEEE 802.3
- Wi-Fi / IEEE 802.11
- PPP (Point-to-Point Protocol)
- DSL, ISDN, Frame Relay
- Fiber-optic / SONET standards

Network (Internet) Layer

Functions & Services

- Logical addressing: assigns and uses IP addresses so packets can be routed across multiple interconnected networks
- Routing: determines the best path for a packet to travel from source to destination via routers
- Fragmentation: splits packets to fit the maximum transmission unit (MTU) of the underlying network, and reassembles them at the destination
- Provides host-to-host (source-to-destination) delivery of individual packets, independent of the physical path taken

Key Protocols

- IP — Internet Protocol (IPv4 / IPv6)
- ARP — Address Resolution Protocol (IP → MAC)
- RARP — Reverse ARP (MAC → IP, largely obsolete)
- ICMP — error reporting & diagnostics (e.g. ping)
- IGMP — Internet Group Management Protocol



QUICK CHECK

Quiz 2 — TCP/IP Layers & Addressing

Q1. Which protocol resolves a known IP address to its corresponding MAC address?

A) RARP B) ICMP C) ARP D) IGMP

Answer: C) ARP

Q2. In the OSI-to-TCP/IP mapping, the OSI Session, Presentation, and Application layers correspond to which single TCP/IP layer?

A) Transport B) Network C) Application D) Data Link

Answer: C) Application

Transport Layer

Functions & Services

- Service-point (port) addressing: delivers data to the correct process/application on the destination host
- Segmentation & reassembly: breaks a message into transmittable units, numbered for correct reassembly
- Connection control: TCP is connection-oriented with setup/teardown; UDP is connectionless and lightweight
- End-to-end flow control and error control, independent of what happens at lower layers
- Choice of protocol here is a key design decision — reliability (TCP) vs. speed (UDP)

Key Protocols

- TCP — Transmission Control Protocol (reliable)
- UDP — User Datagram Protocol (fast, connectionless)
- SCTP — Stream Control Transmission Protocol

Application Layer

Functions & Services

- Combines the roles of the OSI Session, Presentation, and Application layers into one layer
- Provides the interface through which user applications access network services
- Handles data translation, encoding, encryption, and session management within the application itself or its libraries
- Supports services such as web browsing, email, file transfer, name resolution, and remote access

Key Protocols

- HTTP / HTTPS — Web browsing
- FTP, TFTP — File transfer
- SMTP, POP3, IMAP — Email
- DNS — Domain Name System
- TELNET, SSH — Remote login
- SNMP — Network management
- DHCP — Dynamic address assignment

Addressing Across the TCP/IP Layers

Different layers identify "who" using different kinds of addresses — each solving a different part of the delivery problem.

Physical Address

Data Link Layer

MAC address — identifies a device on one local link (e.g. 00:1A:2B:3C:4D:5E)

Logical Address

Network Layer

IP address — identifies a host across the entire internetwork (e.g. 192.168.1.10)

Port Address

Transport Layer

Port number — identifies a specific process/application on the host (e.g. port 443)

Specific Address

Application Layer

Human-friendly address — e.g. a URL or an email address, resolved to the above



QUICK CHECK

Quiz 3 — Application Layer & Addressing

Q1. Which type of address identifies a specific process or application running on a host?

A) Physical address B) Logical address C) Port address D) Specific address

Answer: C) Port address

Q2. Which protocol would you choose for a live video call where speed matters more than guaranteed delivery?

A) TCP B) UDP C) FTP D) SCTP with full reliability

Answer: B) UDP

TCP/IP Layers at a Glance

#	Layer	Example Protocols	Primary Role
5	Application	HTTP, FTP, SMTP, DNS, SSH	User-facing network services
4	Transport	TCP, UDP, SCTP	End-to-end delivery between processes
3	Network	IP, ARP, ICMP, IGMP	Logical addressing & routing
2	Data Link	Ethernet, Wi-Fi, PPP	Framing & MAC addressing
1	Physical	Ethernet PHY, DSL, fiber	Raw bit transmission

Lecture Summary — Key Takeaways

1

TCP/IP is a protocol suite, named after its two foundational protocols: TCP and IP

2

It was designed for robustness, heterogeneity, and distributed control — motivated by ARPANET's needs

3

Cerf & Kahn's 1974 paper and the 1983 "Flag Day" switch from NCP are the key milestones in its history

4

TCP/IP became the global standard over OSI due to availability, simplicity, and real-world momentum

5

The TCP/IP suite has 5 layers (or 4, in the simplified model): Physical, Data Link, Network, Transport, Application

6

Each layer uses a distinct address type — physical (MAC), logical (IP), port, and specific (application-level)



EXAM FOCUS

Important Questions for the Exam

Q1

What is TCP/IP? Why is it described as a "protocol suite" rather than a single protocol?

Q2

List the design goals that shaped the development of TCP/IP.

Q3

Trace the key historical milestones from ARPANET/NCP to modern TCP/IP.

Q4

Why did TCP split into TCP and IP in 1978? What did each protocol take on?

Q5

Explain the significance of the 1983 "Flag Day" in Internet history.

Q6

Why did TCP/IP become the global standard instead of the OSI protocol stack?

Q7

Draw the TCP/IP protocol suite and map its layers to the OSI model.

Q8

Explain the four types of addresses used across the TCP/IP layers.

Interview Questions You May Be Asked

- 1 Can you explain the difference between TCP and IP, in your own words?
- 2 Why does the Internet use TCP/IP instead of the OSI protocol stack?
- 3 What is the role of a router in the TCP/IP model, and at which layer does it operate?
- 4 How does ARP work, and why is it needed alongside IP addressing?
- 5 When would you choose UDP over TCP for an application, and why?
- 6 How would you explain the four types of network addressing to a non-technical colleague?